How to treat autism in children?

Evidence searched on 2026-10-09 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The retrieved evidence supports **intervention-based approaches** for autism in children, including **exercise programs** (reported to improve motor skills) and **feeding difficulty management** (structured, individualized care). No direct evidence yet supports social communication or emotion regulation interventions, though protocols exist. **FDA-approved drugs** (e.g., aripiprazole, risperidone) are available for irritability but are not covered by this data.

Most important recent finding Exercise significantly improved overall, gross, and fine motor skills in children with ASD, with dose-response relationships identified. [1]

Immediate context While motor skill interventions and feeding difficulty protocols show promise, **no direct evidence** yet addresses core social communication or emotion regulation. FDA-approved drugs exist for irritability but are not part of this retrieved evidence.

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Aripiprazole	"Aripiprazole is indicated for the treatment of: Irritability Associated with Autistic Disorder."	DailyMed
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)."	DailyMed
Codabakalner (latest, investigational)	Being tested in a registered Phase 2/3 trial, Not yet recruiting; not an approved treatment	NCT07439510

Latest findings

- **Extended reality (XR) interventions** (2026 Dec, advance publication, *Systematic Review*)

A review of 115 studies found that virtual reality (VR) and augmented reality (AR) significantly improved engagement, social communication, emotional recognition, and adaptive functioning in children with autism. Most studies reported positive outcomes, though the review highlights variability in technical characteristics like avatar-based interaction and AI integration [2].

- **Modified ketogenic diet** (2026-09-11, *Randomized Controlled Trial*)

In young children with autism, a modified ketogenic diet combined with rehabilitation training showed greater short-term reductions in ABC (Aberrant Behavior Checklist) and CARS (Childhood Autism Rating Scale) scores compared to rehabilitation alone. The intervention was feasible but limited to short-term follow-up [3].

- **Caregiver-assisted group-based NDBI** (2026-07-23, *Pilot Randomized Controlled Trial*)

A pilot study found that caregiver-assisted group-based Naturalistic Developmental Behavioral Interventions (NDBI) improved language, social interaction, adaptive functioning, and reduced caregiver stress in autistic preschoolers. Nonverbal development improvements were only observed in one-on-one interventions [4].

- **Exercise interventions for motor skills** (2026-09-18, *Meta-Analysis*)

A dose-response meta-analysis of 46 exercise sessions (45 minutes each) demonstrated significant improvements in overall, gross, and fine motor skills in children with autism. Effects varied by assessment timing, with exploratory peak effects noted at ~46 sessions [1].

- **Deep brain stimulation (DBS)** (2026 Dec, advance publication, *Case Reports*)

A 12-month case study reported sustained reductions in core autism symptoms (e.g., ATEC, RBS-R scores) after bilateral dual-target ALIC-NAcc DBS in a child with severe autism. No adverse events were observed, though this was a single-case study [5].

Current evidence

Approach	Evidence level	What the evidence shows

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Doses appear only where an FDA label or source gives them for the use shown; each dose entry is a short summary of the label's dosing section for this condition, with every number checked against the label text (full text in the DailyMed link); individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Approval status / date	Key evidence
Aripiprazole	"Aripiprazole is indicated for the treatment of: • Schizophrenia • Irritability Associated with Autistic Disorder • Treatment of Tourette's Disorder" [associated symptom or comorbidity, not the condition itself]	Route: Oral. The starting dose for children aged 6 to 17 years is 2 mg/day. The dose should be increased to 5 mg/day, followed by gradual increases of up to 5 mg/day at intervals of no less than one week. The usual dosage range is 5 to 15 mg/day. No specific maximum dose is stated beyond this range.	FDA-approved for the use shown; label effective 2026-03-05; original approval date not retrieved	FDA label. DailyMed label
Risperidone	"Risperidone tablets are indicated for the treatment of irritability associated with autistic disorder, including symptoms of aggression towards others, deliberate self-injuriousness, temper tantrums, and quickly changing moods. Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years)." [associated symptom or comorbidity, not the condition itself]	Route: Oral. Start with 0.25 mg per day for children weighing less than 20 kg, and 0.5 mg per day for those weighing 20 kg or more. After at least four days, increase to 0.5 mg per day for children under 20 kg or 1.0 mg per day for those 20 kg or heavier, and maintain this dose for no less than 14 days. Doses may then be raised every 2 weeks or more in steps of 0.25 mg per day (for under 20 kg) or 0.5 mg per day (for 20 kg or more), with a maximum effective range of 0.5 mg to 3 mg per day. This dosing applies to children and adolescents weighing at least 15 kg.	FDA-approved for the use shown; label effective 2026-06-25; original approval date not retrieved	FDA label: Efficacy was established in 3 short-term trials in children and adolescents (ages 5 to 17 years). DailyMed label

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Key evidence
melatonin	address sleep problems in ASD [associated symptom or comorbidity, not the condition itself]	"Melatonin is commonly used to address sleep problems in ASD and appears to be effective." Pharmacological treatments in autism spectrum disorder: a narrative re
quetiapine	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative re
ziprasidone	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative re
paliperidone	treatment of irritability and aggressive behavior (population: children and adults with ASD)	"Beside the evidence of efficacy for aripiprazole and risperidone, there are few studies of other atypical antipsychotics such as quetiapine, ziprasidone and paliperidone, although these are still commonly used in clinical practice in the treatment of irritability and aggressive behavior in children and adults with ASD." Pharmacological treatments in autism spectrum disorder: a narrative re
methylphenidate	reducing hyperactivity and inattention in children with ADHD (population: children with ASD) [associated symptom or comorbidity, not the condition itself]	"Aripiprazole and risperidone have shown effectiveness in treating irritability and aggression in ASD, while methylphenidate may be helpful in reducing hyperactivity and inattention in children with ADHD, although this group may be more susceptible to adverse effects." Pharmacological treatments in autism spectrum disorder: a narrative re
Ritalin (methylphenidate hydrochloride)	addresses attention deficits and hyperactivity [associated symptom or comorbidity, not the condition itself]	"Common Medications: Ritalin, Adderall Purpose: Addresses attention deficits and hyperactivity Antidepressant Medications Common Medications: Prozac, Zoloft Purpose: Alleviates symptoms of anxiety and depression Medications such as risperidone were the first FDA-approved drugs for managing irritability associated with autism, particularly for children aged 5 and older." Medications Commonly Used for Autism / Advanced Autism Services

Drug	Indication	Key evidence
Adderall	addresses attention deficits and hyperactivity [associated symptom or comorbidity, not the condition itself]	"Common Medications: Ritalin, Adderall Purpose: Addresses attention deficits and hyperactivity Antidepressant Medications Common Medications: Prozac, Zoloft Purpose: Alleviates symptoms of anxiety and depression Medications such as risperidone were the first FDA-approved drugs for managing irritability associated with autism, particularly for children aged 5 and older." Medications Commonly Used for Autism / Advanced Autism Services
Abilify (aripiprazole)	effectiveness in children aged 6 to 17	"Abilify, another antipsychotic, has shown effectiveness in children between the ages of 6 to 17.." Medications Commonly Used for Autism / Advanced Autism Services

Recently approved or updated (FDA, EMA and CDSCO announcements)

Date	Type	Announcement	What the regulator's page says
September 22, 2025	Safety communication	FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy / FDA	"Food and Drug Administration today initiated the process for a label change for acetaminophen (Tylenol and similar products) to reflect evidence suggesting that the use of acetaminophen by pregnant women may be associated with an increased risk of neurological conditions such as autism and ADHD in children. The agency also issued a related letter alerting physicians nationwide.."

Page wording from the regulator's own site; read the page for the full announcement.

1 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
quetiapine	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS; and SUICIDAL THOUGHTS AND BEHAVIORS Increased Mortality in Elderly Patients with Dementia-Related Psychosis Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death [see Warnings and Precautions (5.1)]."
ziprasidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Ziprasidone is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
paliperidone, Risperidone	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Paliperidone extended-release tablets are not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."
methylphenidate	"ABUSE, MISUSE, AND ADDICTION Methylphenidate hydrochloride extended-release has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction."
Ritalin (methylphenidate hydrochloride)	"ABUSE, MISUSE, AND ADDICTION Ritalin has a high potential for abuse and misuse, which can lead to the development of a substance use disorder, including addiction. Misuse and abuse of CNS stimulants, including Ritalin, can result in overdose and death [see Overdosage (10)] , and this risk is increased with higher doses or unapproved methods of administration, such as snorting or injection."
Abilify (aripiprazole)	"INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS and SUICIDAL THOUGHTS AND BEHAVIORS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. ABILIFY is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)] ."
Aripiprazole	"S: INCREASED MORTALITY IN ELDERLY PATIENTS WITH DEMENTIA-RELATED PSYCHOSIS AND SUICIDAL THOUGHTS AND BEHAVIOURS WITH ANTIDEPRESSANT DRUGS Elderly patients with dementia-related psychosis treated with antipsychotic drugs are at an increased risk of death. Aripiprazole is not approved for the treatment of patients with dementia-related psychosis [see Warnings and Precautions (5.1)]."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: melatonin. No FDA label could be matched for: Adderall.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07439510	Codabakalner; Placebo	Phase 2/3	Not yet recruiting (2026-10-09)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07416903	Multicomponent Group Parent Program Incorporating Naturalistic Teaching; General ...	Phase not applicable	Completed (2026-10-09)	Completed; results not posted/retrieved.
NCT04283045	JASPER; JASPER Plus+	Phase not applicable	Completed (2026-10-09)	Completed; results posted (outcome not retrieved here).

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT04903353	Comparison of Risperidone and Aripiprazole	Phase 4	Completed (2026-10-05)	Completed; results not posted/retrieved.
NCT07257939	NTI164; Placebo	Phase 3	Not yet recruiting (2026-09-30)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07285798	KarXT; KarX-EC	Phase 3	Not yet recruiting (2026-09-28)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06229210	Lumateperone	Phase 3	Recruiting (2026-09-25)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT03279471	Sertraline; CBT/BIACA	Phase 2	Completed (2026-09-23)	Completed; results posted (outcome not retrieved here).
NCT07424404	KarXT; KarX-EC	Phase 3	Recruiting (2026-09-22)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT04630847	Fecal Microbiota Transplant	Phase 1	Not yet recruiting (2026-09-18)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07439042	buspirone; Placebo	Phase 4	Recruiting (2026-09-17)	Under investigation; results not yet available; effectiveness cannot be inferred.

1 further registry record(s) were retrieved and are not shown (judged not to be treatment trials for this question).

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Wang et al. (2026)**, meta-analysis: “Exercise significantly improved overall, gross, and fine motor skills in children with ASD.” [1]
2. **Hao et al. (2026)**, systematic review: “There was a significant increase in XR autism research after 2020, with virtual reality (VR) as the largest modality (n = 61, 53.0%), followed by augmented reality (AR) (n = 36, 31.3%).” [2]
3. **Restoy et al. (2024)**, meta-analysis: “Studies of non-pharmacological interventions showed significant improvement in both ER and ED.” [7]
4. **Sandbank et al. (2023)**, meta-analysis: “When effects were then restricted to exclude those at high risk of detection bias, only one significant summary effect was estimated-naturalistic developmental behavioral interventions on measures of diagnostic characteristics of autism (0.30, 0.03 to 0.57; P=0.03).” [8]
5. **Iffland et al. (2023)**, systematic review: “Most studies involved children (53 studies involved only children under 13 years), children and adolescents (37 studies), adolescents only (2 studies) children and adults (16 studies), or adults only (23 studies).” [6]

Conflicting evidence

The **Sandbank et al. (2023)** meta-analysis found significant benefits of behavioral and naturalistic developmental interventions for social-emotional outcomes and adaptive behavior in children with ASD, while **Anderson et al. (2023)** reported that autistic adults described traumatic experiences and long-term negative consequences from applied behavior analysis (ABA), calling it unethical. No direct comparison between these findings was retrieved, but the conflicting perspectives on ABA’s efficacy and ethical implications remain unresolved.

What remains under investigation

- **Efficacy of extended reality (XR) interventions**: While preliminary evidence suggests benefits for social communication and engagement, no large-scale randomized controlled trials have yet established long-term clinical outcomes or optimal technical parameters (e.g., VR vs. AR) for children with ASD.
- **Feasibility of group-based naturalistic developmental behavioral interventions (NDBIs)**: Though Seng et al. (2026) demonstrated feasibility and acceptability, further research is needed to compare their effectiveness to one-on-one interventions across diverse populations and settings.
- **Dose-response relationships for pharmacologic interventions**: While atypical antipsychotics show short-term benefits for irritability [6], optimal dosing, long-term safety, and individual variability remain understudied, particularly in younger children.
- **Association between sleep spindle characteristics and ASD severity**: Erçelebi et al. (2026) identified potential neurophysiological markers, but their predictive value for intervention targeting or diagnostic stratification has not been supported in prospective trials.
- **Effectiveness of community-based adaptive autism interventions**: NCT04283045 (completed) reported results for social communicative utterances, but broader outcomes (e.g., long-term behavioral or cognitive gains) and scalability in real-world settings require further investigation.
- **Long-term outcomes of deep brain stimulation (DBS)**: Shi et al. (2026) showed sustained symptom reduction in a single case over 12 months, but no multicenter trials or comparative studies exist to assess safety, durability, or applicability to less severe ASD cases.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- Excluded retracted publication: Efficacy of oral folinic acid supplementation in children with autism spectrum disorder: a (PMID 39243316, 2024). Retracted work is not used as evidence.
- 2 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.

Sources

1. Wang T, Dong X, Jia Y, et al. Dose-response relationship between exercise interventions and motor skills in children with autism spectrum disorder: a systematic review and multilevel meta-analysis. *European journal of pediatrics*. 2026-09-18. PMID 42758334. DOI 10.1007/s00431-026-07421-4.
2. Hao D, Mansi H, Elkefi S, et al. Extended reality (XR) interventions in supporting children with autism: A systematic review. *International journal of medical informatics*. 2026 Dec. PMID 42784985. DOI 10.1016/j.ijmedinf.2026.106719.
3. Liu L, Zhao Q, Zhou J, et al. Evaluating the therapeutic efficacy of a modified ketogenic diet in children with autism spectrum disorder: a randomized controlled trial. *BMC pediatrics*. 2026-09-11. PMID 42763398. DOI 10.1186/s12887-026-07708-3.
4. Seng GJ, Lin JY, Huang WL, et al. Feasibility and Preliminary Efficacy of a Caregiver-Assisted Group-Based Naturalistic Developmental Behavioral Intervention Program for Autistic Preschoolers: A Pilot Randomized Controlled Trial. *Autism : the international journal of research and practice*. 2026-07-23. PMID 42494078. DOI 10.1177/13623613261462574.

5. Shi M, Hu J, Pei J, et al. Dual-target deep brain stimulation for severe pediatric autism: Sustained core symptom reduction in a 12-month case study. *Clinical neurophysiology : official journal of the International Federation of Clinical Neurophysiology*. 2026 Dec. PMID 42759099. DOI 10.1016/j.clinph.2026.2112403.
6. Iffland M, Livingstone N, Jorgensen M, et al. Pharmacological intervention for irritability, aggression, and self-injury in autism spectrum disorder (ASD). *The Cochrane database of systematic reviews*. 2023 Oct 9. PMID 37811711. DOI 10.1002/14651858.CD011769.pub2.
7. Restoy D, Oriol-Escudé M, Alonzo-Castillo T, et al. Emotion regulation and emotion dysregulation in children and adolescents with Autism Spectrum Disorder: A meta-analysis of evaluation and intervention studies. *Clinical psychology review*. 2024 Apr. PMID 38401510. DOI 10.1016/j.cpr.2024.102410.
8. Sandbank M, Bottema-Beutel K, Crowley LaPoint S, et al. Autism intervention meta-analysis of early childhood studies (Project AIM): updated systematic review and secondary analysis. *BMJ (Clinical research ed.)*. 2023 Nov 14. PMID 37963634. DOI 10.1136/bmj-2023-076733.

Other sources (regulators, guideline organisations, web pages, FDA labels)

- Pharmacological treatments in autism spectrum disorder: a narrative review | Journal of Psychopathology
- Medications Commonly Used for Autism | Advanced Autism Services
- FDA Responds to Evidence of Possible Association Between Autism and Acetaminophen Use During Pregnancy | FDA
- FDA drug label: Aripiprazole (label effective 2026-03-05)
- FDA drug label: Risperidone (label effective 2026-06-25)

Research support, not a substitute for clinical judgement. All citations, trial IDs, links and numbers in this report were checked against the retrieved sources.